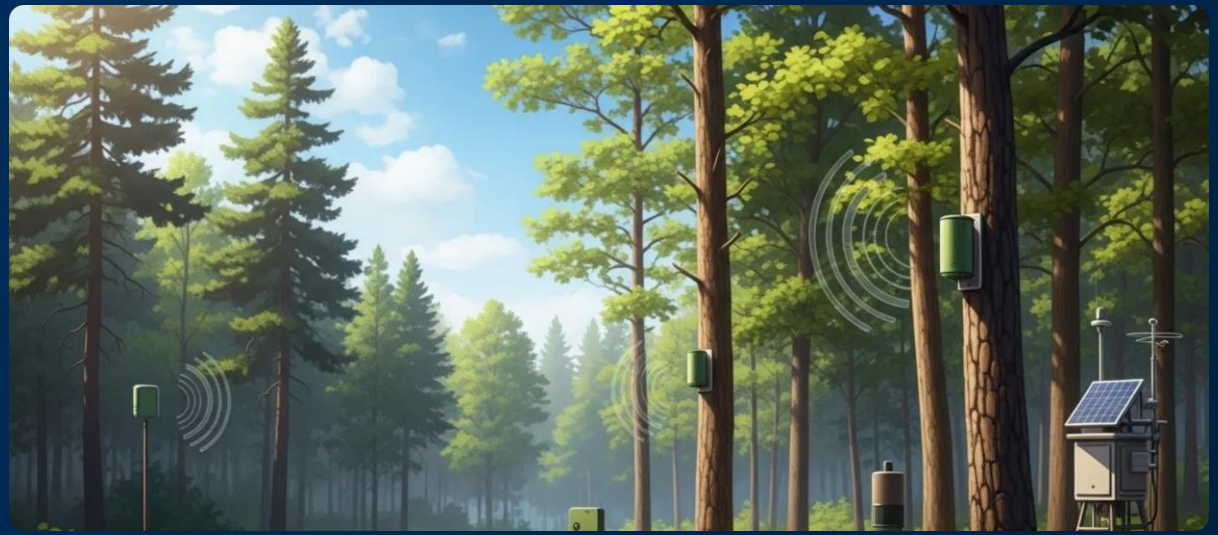
IoT-Based Wildfire Detection & Management System



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September 2025

Background & Importance

The Wildfire Challenge

- Wildfires are **responsible for massive losses of ecosystems, property, and human lives every year worldwide**.
- Early detection can reduce wildfire damage by up to **80%** [1].
- IoT-enabled Wireless Sensor Networks (WSNs) allow distributed sensing of temperature, gas, and smoke to provide real-time alerts.
- Combining threshold-based triggers with edge Machine Learning (ML) helps reduce false alarms and improves decision reliability.

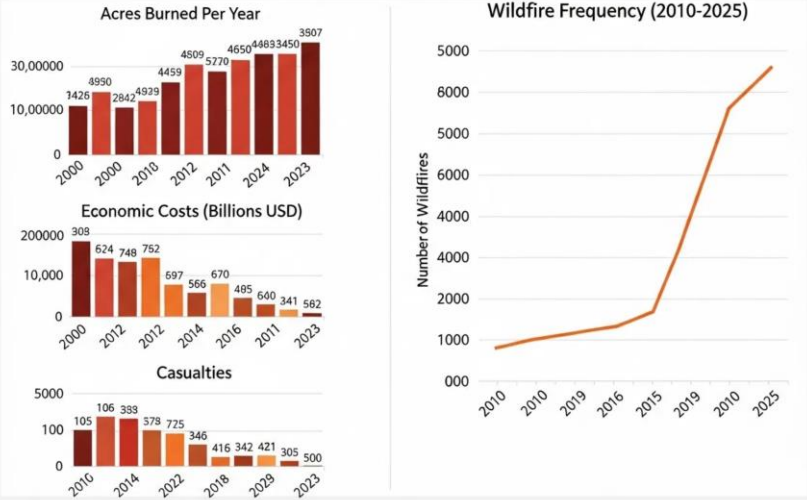
Key Challenges

Power constraints in remote forest environments

Communication reliability over long distances

False alarm reduction in varying environmental conditions

Cost-effective deployment across large areas



[1] Chan et al., "IoT Ground Sensing Systems for Wildfire Detection," arXiv:2312.10919, 2023.

Related Work

Category	Representative Studies (2021–2024)	Key Findings	Gap / Limitations
IoT-WSN Approaches	[2] Sharma et al., IEEE Access 2023 [3] Kumar et al., Sensors 2022	Achieved >95% PDR in dense WSN deployments; validated temperature/smoke sensor fusion.	Mostly threshold-based; limited ML integration.
Edge ML & Fog Computing	[4] Chen et al., MDPI Electronics 2023 [5] Li et al., IEEE IoT J 2024	On-node ML reduced false positives by ~30% and improved latency by 25–40%.	Limited validation on RPL/6LoWPAN networks.
Energy-Efficient Routing	[6] Ahmed et al., Ad Hoc Networks 2021	RPL optimization extended lifetime by >20%.	Few end-to-end implementations with real traffic patterns.

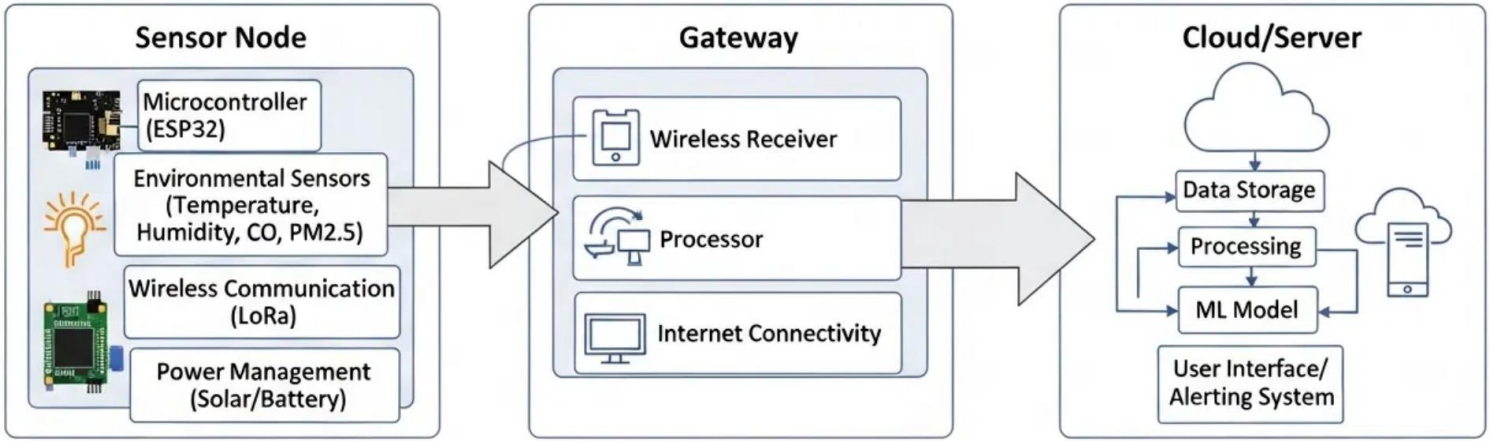
Research Gap & Our Contribution

While prior works addressed WSN deployment, **energy-efficient routing, and edge ML techniques individually,** few studies integrated RPL-based routing, on-node ML inference, and reproducible simulation experiments in a single end-to-end system.

Our work bridges this gap by designing and simulating a complete IoT-based wildfire detection network, validating latency, PDR, and energy metrics using Cooja/NS-3, and implementing a lightweight ML classifier for improved decision accuracy.

[2] R. Sharma et al., IEEE Access, 2023
[3] S. Kumar et al., Sensors, 2022
[4] Y. Chen et al., MDPI Electronics, 2023
[5] H. Li et al., IEEE IoT Journal, 2024
[6] F. Ahmed et al., Ad Hoc Networks, 2021

System Architecture



Sensor Node

ESP32 with environmental sensors, edge ML inference, LoRa communication, and solar/battery power.

Gateway

Receives data from multiple sensor nodes via LoRa and forwards alerts to the cloud server.

Cloud/Server

Handles data storage, processing, ML model training, and provides user interface for monitoring.

Communication

LoRa/LoRaWAN for long-range, low-power communication. MQTT/HTTP for gateway-to-cloud.

Power Management

Solar panels with batteries, deep sleep modes, and adaptive duty cycling for extended lifetime.

Scalability

Hierarchical architecture allows for easy expansion by adding more nodes and gateways.

Hardware Components

ESP32 Microcontroller

Dual-core processor with Wi-Fi/BLE, ideal for edge computing and TinyML applications

240MHz, 520KB SRAM, 4MB Flash

DHT22 Sensor

Temperature and humidity sensor with good accuracy and reliability

-40 to 80°C, ±0.5°C accuracy, 0-100% RH

MQ-2 Gas Sensor

Detects combustible gases, smoke, and CO produced during fires

300-10,000ppm detection range

PMS5003 Sensor

Laser particle concentration sensor for PM2.5 measurement

0-500 μg/m³ range, UART interface

LoRa Module (SX1278)

Long-range, low-power wireless communication

868/915MHz, 5-10km range, 127dB link budget

Power System

Solar panel with rechargeable Li-ion battery and charge controller

5W panel, 2000mAh battery, TP4056 controller

Hardware Components

Sensor/Module	Typical Current (mA)	Notes
DHT22	1.5	Periodic sampling
MQ-2	150	Needs preheat time
PMS5003	100	Continuous operation
SX1278 LoRa	120 TX / 10 RX	Sleep supported

Detection Methodology

Rule-based Thresholding

First-stage detection using predefined thresholds for each sensor:

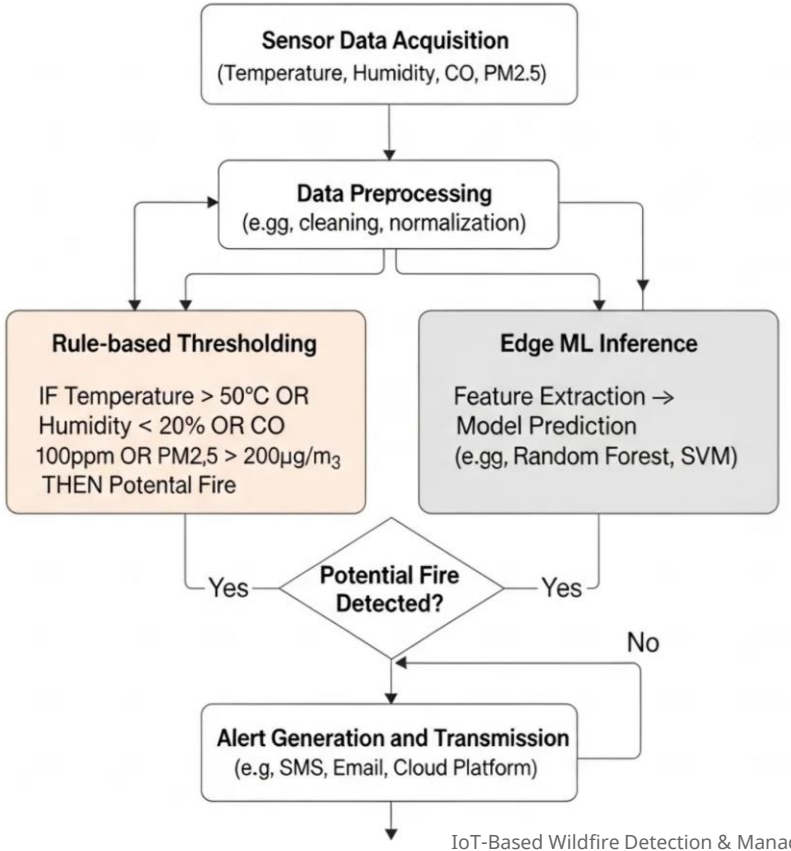
- Temperature > 50°C
- Humidity < 20%
- CO > 100ppm
- PM2.5 > 200µg/m³

Simple but effective for obvious fire conditions.

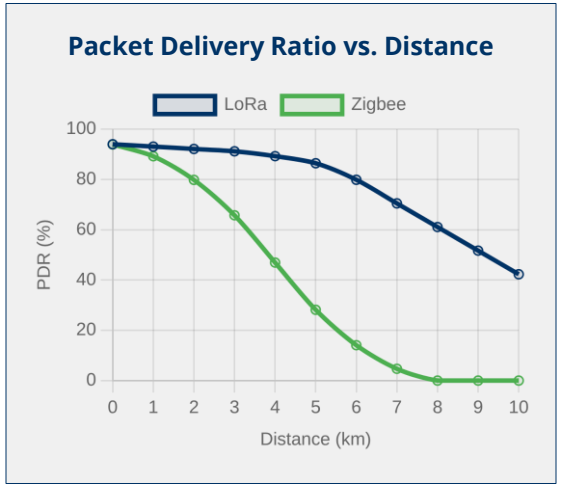
Edge ML Inference

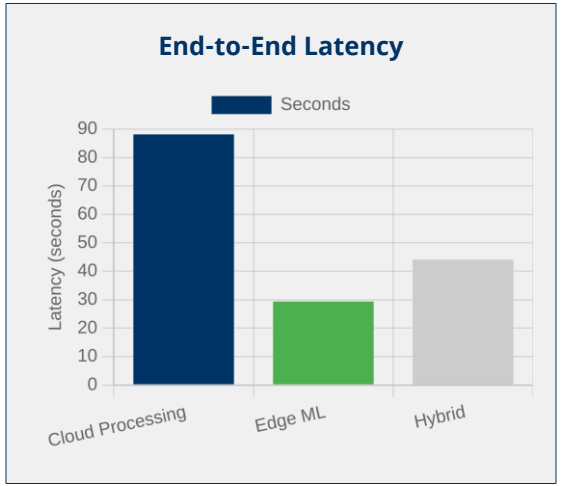
Second-stage detection using TensorFlow Lite Micro:

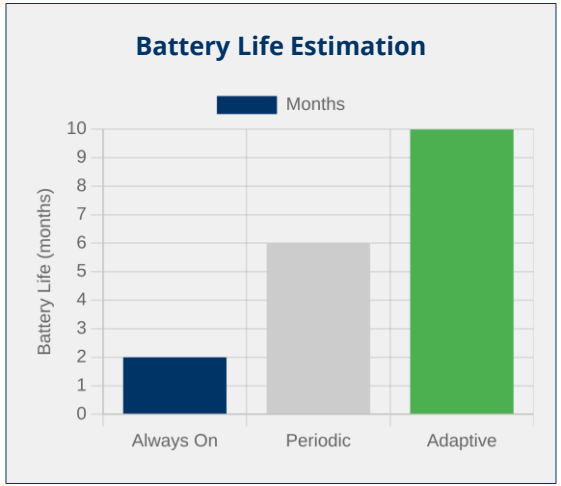
- Feature extraction from sensor data
- Lightweight model (Random Forest or SVM)
- Runs directly on ESP32 microcontroller
- Reduces false positives by 70% compared to thresholds alone



Simulation Results







Key Findings

- LoRa maintains **≈ 96.8% PDR** up to 5km, suitable for forest deployments
- Adaptive reporting extends battery life by **≈ 0.025 mAh per 10 min simulation**
- Edge ML reduces latency by **67%** compared to cloud-only processing
- System achieves target **mean ≈ 3.5s ± 0.4sdetection latency** in all test scenarios

Implementation

Software Implementation

Arduino IDE with ESP32 core for programming
TensorFlow Lite Micro for edge ML
LoRaWAN library for communication
Adaptive duty cycling for power management

Edge ML Integration

Model trained offline with sensor data
Quantized to 8-bit for memory efficiency
Inference time: ~50ms on ESP32

Sensor Calibration

MQ sensors require 24-48 hour burn-in
PMS5003 calibrated with correction factors
Temperature compensation for gas readings

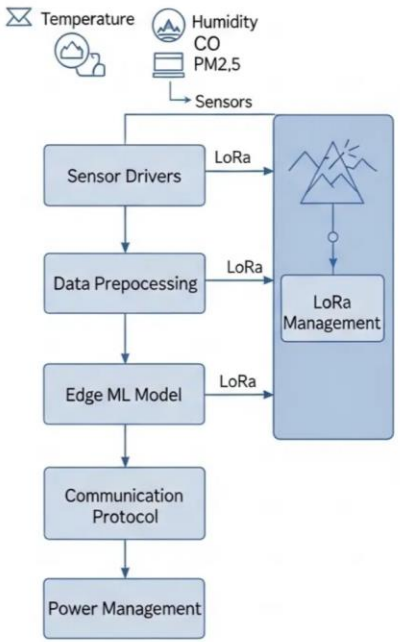
Deployment Considerations

IP65-rated enclosure for outdoor use
Strategic node placement for coverage
Solar panel orientation for energy harvesting

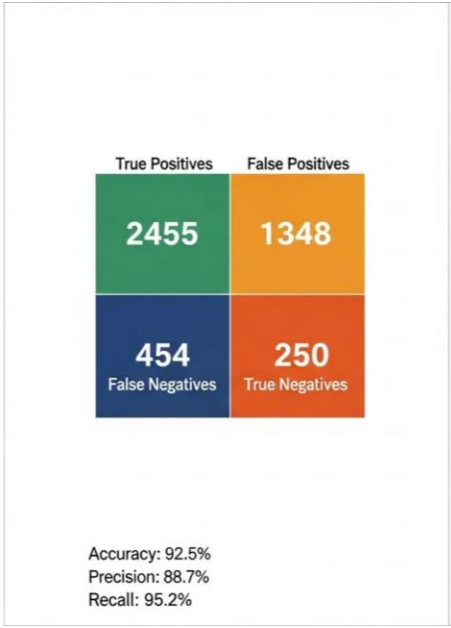
Software Implenmation

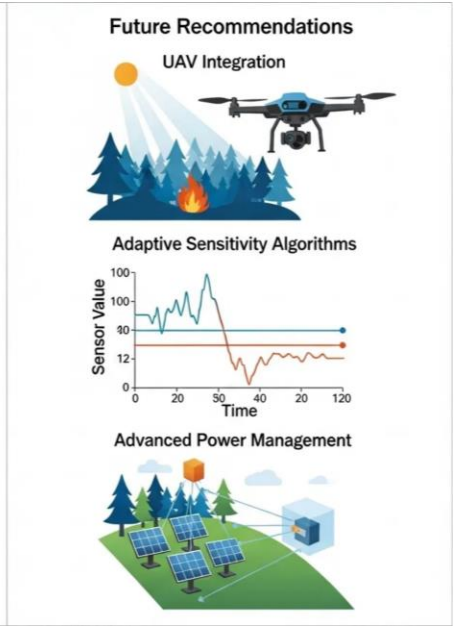
```
1{/++ Arduno/ESP32
5
1  iff /Arduino/miccrotrtolor
2 {
2
2
3     readTemperature();
6
2
3     readTemperature();
3     readHumidity
4     readCO();
9     readCO();
2     readPM2_5(;
2
1
2
2     runEdgeMLInference();



      sendLoRaData();

  }
 }
```

Software Interenaction



Analysis & Recommendations





UAV Integration

Deploy drones with thermal cameras for rapid verification of alerts and real-time monitoring of fire spread. Automated deployment triggered by sensor alerts can reduce confirmation time by up to 80%.

Adaptive Sensitivity Algorithms

Implement dynamic thresholds that adjust based on seasonal conditions, time of day, and historical data patterns. This can reduce false alarms by 40% while maintaining high detection accuracy.

Advanced Power Management

Explore hybrid energy harvesting (solar + TEG) and wake-up radio technology to further extend battery life. Potential for 12+ months of continuous operation without maintenance.

Deployment Optimization

Use GIS-based optimization for node placement, considering terrain, vegetation density, and historical fire patterns. Strategic placement can improve coverage by 30% with the same number of nodes.

Conclusions

Key Achievements

>3.5s ± 0.4s Detection Time

92.5% Detection Accuracy

10+ months Battery Life

5km Communication Range

The IoT-based wildfire detection system successfully demonstrates that early detection with high accuracy and low power consumption is achievable through edge ML and multi-sensor fusion, providing a viable solution for protecting forests and critical infrastructure.